STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 6: System of Particles and Rotational Motion

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Mass-weighted mean position.

rF sintheta.

kg m^2.

Iomega^2.

Net external torque is zero.

v=romega.

F=0 and tau=0.

I=I_cm+Md^2.

3 m.

2 N m.

24 J.

20 rad/s.

3 rad/s^2.

Constant.

3 m.

5 N m.

I=.5 kg m^2; K=25 J.

2 rad/s^2; 24 rad.

3 m; 2 m/s^2.

omega=7.5 rad/s; K increases 18 to 45 J because skater does work.